Natural Language Processing Fundamentals

Synthetic educational document created for RAG development and evaluation practice. It is intended as a controlled development dataset, not as an authoritative source.

1. Introduction to NLP

Natural language processing, or NLP, is the field of using computational methods to process and understand human language. NLP tasks include text classification, sentiment analysis, named entity recognition, information extraction, question answering, summarization, translation, and text generation.

2. Text Preprocessing

Traditional NLP pipelines often begin with preprocessing. Tokenization divides text into smaller units such as words or subwords. Lowercasing can reduce vocabulary size, although it may remove useful information in some tasks. Stop-word removal eliminates selected common words, while stemming and lemmatization attempt to reduce words to simpler forms. Modern transformer systems often use subword tokenization rather than extensive manual preprocessing.

3. Bag of Words and TF-IDF

The bag-of-words representation describes a document using word occurrence information while ignoring word order. TF-IDF gives higher weight to terms that are important in a document but less common across the collection. These methods are simple and remain useful as baselines for text classification and information retrieval.

4. Word Embeddings

Word embeddings represent words as dense numerical vectors. Similar words can have related vector representations. Word2Vec and GloVe are classic embedding approaches. Modern NLP systems often use contextual representations, where the representation of a word depends on the surrounding text.

5. Text Classification

Text classification assigns categories to documents or messages. Examples include spam detection, sentiment classification, topic classification, and intent detection. A classification pipeline may combine tokenization, feature extraction or embeddings, a classifier, and an evaluation procedure.

6. Named Entity Recognition

Named entity recognition identifies spans of text that correspond to categories such as people, organizations, locations, dates, or products. For example, in the sentence 'OpenAI was founded in San Francisco', an NER system might identify OpenAI as an organization and San Francisco as a location.

7. Information Retrieval

Information retrieval is the task of finding relevant information from a collection in response to a query. Traditional retrieval can use keyword-based methods such as BM25. Modern retrieval systems can also

use dense embeddings to search for semantically similar passages. Retrieval is a key component of many question-answering and RAG systems.